INCH-POUND

MIL-P-14852B (AR)
AMENDMENT 2
23 August 2000
SUPERSEDING
AMENDMENT 1
22 June 1994

MILITARY SPECIFICATION

PROJECTILE, 105MM, ILLUMINATING,
M314A3, METAL PARTS ASSEMBLY

MIL-P-14852 was inactivated after 9 July 1996 for new design

This amendment forms a part of Military Specification MIL-P-14852B (AR), dated 30 November 1998, and is approved for use by the U.S. Army, Armament Research, Development and Engineering Center and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* Para 2.11

Under specifications, Military delete the following:

MIL-A-2550 - Ammunition, and Special Weapons General Specification for,

MIL-M-6867 - Magnetic Inspection Units

MIL-H-6875 - Heat Treatment of fleets (aircraft) Practice Process for,

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for,

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* Para 2.1.1

Under Standards, Military Delete the following:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by attributes

MIL-STD-410 - Nondestructive Testing Personnel

MIL-STD-1167 - Ammunition Data Card

MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by attributes

MIL-STD-1949 - Inspection Process, Magnetic Particle

Under Standards, MILITARY ADD the following:

MIL-STD-1168 - Lot Numbering of Ammunition and Ammunition Data Card

MIL-STD-1916 - DOD preferred Methods for Acceptance of Product

* Para 2.1.2 Under Drawings, Delete the following:

8837835 - Pallet for fixed and semi-fixed shell

* Para 2.2

Add the following under American Society for Testing and Materials.

ASTM Method E103 - Rapid Indentation of Hardness Testing Of Metalic Materials

ASTM Method E1444 - Standard Practice for Magnetic Particle Inspection

ASTM D3951 - Standard Practice for Commercial Packaging

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Change the address for ASTM From:

1916 Race Street, Philadephia, PA 19103

To:

100 Barr Harbor Drive, West Conshohocken, PA

* Add the following new paragraph:

Aerospace Industries Association of America, Inc. (AIA/NAS)

NAS-410 NAS Certification and Qualification of Non-destructive Test Personnel

(Application for copies should be addressed to the Aerospace Industries Association of America, Inc. (AIA/NAS), 1250 Eye Street, NW, Suite 1100, Washington, D.C. 20005)

* Add the following new paragraph:

American National Standard Institute (ANSI)

B-46.1 Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018).

PAGE 4

* 3.2 Delete last sentence in the paragraph.

PAGE 5

3.4 Delete paragraph in its entirety and substitute with:

“3.4 Rotating band seating.

3.4.1 Rotating band cleanliness. Prior to banding, the band seat shall be clean and free of oil, dirt, grease, rust or other foreign matter. Shot or sand blasting is not permitted for cleaning the band seat after knurling.

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3.4.2 Band blank – (gilding metal). The rotating band blank shall be in the form of a solid ring and shall be annealed to comply with the applicable specification. Annealing prior to banding shall not be necessary in the case of hot banding in excess of 800 degrees F.

3.4.3 Application of band blank. The rotating band blank shall be pressed into the band seat after final heat treatment of the projectile body. The band blank shall be applied in such a manner as not to distort the projectile body.

3.4.4 Sidewall contact. The band blank must contact the sidewall of the band seat for a minimum of eight-tenths of the band seat depth, measured from the outside surface of the projectile body to the rotating band undercut surface.

3.4.5 Sidewall gap. There shall be no gap between the rotating band and the band seat sidewalls at the outside surface of the projectile body after the rotating band has been machined to final dimensions.

3.4.6 Rotating band clearance. The diametrical clearance between the rotating band and the band seat shall not exceed .006 inch, when the destructive method of testing is used, or .009 inch when the non-destructive method of testing is used."

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3.6, Delete paragraph in its entirety and substitute the following:

"3.6 Proving ground functional tests. The projectile assembly shall reveal no evidence of failure when ballistically tested at excess pressure (see 4.5.5.1). Metal part failure shall be defined as a breakup either in the bore of the weapon or in flight."

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* 3.10.3 Delete in its entirety and substitute the following:

3.10.3 Threads. Threads shall be full and undamaged for the entire minimum length or depth as specified on the applicable drawing.

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* 4.3.1 Change Quantities for Part No. 10534900 from 20 to 27.

* 4.3.2 Delete MIL-A-48078 and substitute MIL-STD-1916.

* 4.3.3 Delete MIL-A-48078 and substitute MIL-STD-1916

PAGE 10

* Table I.

Change quantity for part number 10534900 from 15 to 25.

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* 4.4.1 Delete MIL-A-48078 and substitute MIL-STD-1916.

PAGE 12

* 4.4.2 Delete MIL-A-48078 and substitute MIL-STD-1916.

* 4.4.2a Delete paragraph and substitute with:

Para 4.4.2a Sampling plans. Unless otherwise specified in the classification of Defects and Test Tables, sampling plans will be in accordance with MIL-STD-1916, Verification Level IV for major defects, and Level II for minor defects.

* Pages 13 thru 20, Quality Conformance Inspection Table.

Paragraphs 4.4.2.1 thru 4.4.2.7

In AQL or 100% Column, delete “0.40% and 0.65% AQLs.

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4.4.2.4 Add the following new requirement paragraph to Major Category 105:

“3.4.1”

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4.4.2.5 Change Major Category 104 from “4.4.3.1.2.6” to “4.4.3.1.2.7”.

* Add the following new Major Defect 106:

“106 Sidewall gap 3.4.5 GAGE.”

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* 4.4.3.1.2.7, Delete in its entirety and substitute the following:

“4.4.3.1.2.7 Rotating band. The sample rotating band for testing in accordance with 4.5.1.2.3 for compliance with 3.3.2 shall be provided as part of the sample taken from 4.4.3.2.1.3. The sample size, frequency and rejection criteria is the same as that described in 4.4.3.2.1.3.”

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* 4.4.3.2.1.3 Delete in its entirety and replace with:

“4.4.3.2.1.3 Sidewall contact. One body assembly shall be selected at the beginning of each shift (not to exceed ten hours) from each banding machine and sectioned in accordance with 4.5.2.3 for compliance with 3.4.4 and also 3.3.2 (see 4.4.3.1.2.7). If, for any reason, there is a change in operator, machinery, or material (by lot number) during a shift, another sample shall be taken. Failure to comply with the requirements of the specification shall be cause for rejection of all material processed since the previous sample.”

* 4.4.4 Delete last sentence in its entirety and substitute “See Section 6.3 herein.

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* 4.4.4.2 Delete in its entirety and replace with:

4.4.4.2 Certification of nondestructive test (NDT) personnel.

All personnel operating or calibrating NDT equipment shall be qualified and certified in accordance with NAS-410.

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Para 4.5.1.2.3 Change reference paragraph from “4.4.3.1.2.6” to “4.4.3.1.2.7”.

4.5.2, Delete paragraph in its entirety and replace with:

“4.5.2 Rotating band seating.

4.5.2.1 Clearance (Nondestructive method). The samples provided by 4.4.3.2.1.1 shall be tested for the requirements in 3.4.6 utilizing the equipment and procedures prescribed in the Detroit Testing Machine Manual (see 6.7).

4.5.2.2 Clearance (Destructive method). The body assembly selected in accordance with 4.4.3.2.1.2 for compliance with 3.4.6 need not be processed through the band machining line, although the band should have a light surface cut (.020” max.) to provide a uniform surface for measurement. The diameter of each band shall be measured at three locations, sixty degrees apart, on the points along the cylindrical portion of their length, near the front and rear edges. This results in six (three pairs of two) diametrical measurements per band. The points of measurement shall be marked on the band and the projectile body adjacent thereto in such a manner as to permit proper body and band orientation after removal. The band shall be carefully removed so as to keep deformation to a minimum. After removal, the thickness of the band shall be measured at each of the marked points (a total of twelve measurements) and the diameter of the band seat corresponding to these points shall also be measured. Add to each of the diameters of the band seat the sum of the two corresponding two thicknesses of the band; the results obtained shall then be subtracted from the previously determined diameters over the band.

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4.5.2.3 Sidewall contact. The samples provided in accordance with 4.4.3.2.1.3 shall be sectioned by cutting through two planes normal to the longitudinal axis one-half inch forward of the rotating band. The cylindrical portion containing the rotating band shall be sectioned approximately in half through a plane parallel to its longitudinal axis. One of the two halves shall be used for establishment of mechanical properties of the band; the other section shall be sectioned into two parts, exposing four surfaces for the viewing of the band swaged into the band seat. The exposed surfaces shall be ground, and the metal chip shall be removed prior to measurement for depth of sidewall contact to determine compliance with 3.4.4."

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* 4.5.4.1 Delete MIL-STD-1949 and substitute ASTM E1444.

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* 4.5.4.2 Delete MIL-STD-1949 and substitute ASTM E1444.

* 4.5.4.3 Delete in second sentence "Section 4 of MIL-STD-1949", and substitute "ASTM E1444".

* 4.5.4.3 Delete last sentence and substitute "See paragraph 6.3 and ASTM E1444".

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* 4.5.5 Delete the entire paragraph and substitute the following:

4.5.5 Proving ground function tests.

4.5.5.1 Excess pressure. The metal parts assemblies supplied in accordance with 4.4.3.4 inert loaded and assembled in a complete round, shall be fired in a gun for which the projectile is standard, the tube of which shall have a minimum of 25 percent life remaining based on current gun tube condemnation criteria. A propellant charge adjusted to yield a mean pressure of 42,500 psi with no maximum individual exceeding 44,400 psi shall be used. Photographic equipment shall be set up near the gun muzzle to record the flight of the projectile to determine compliance with 3.6. Recovery and examination of metal parts for any abnormal condition shall be accomplished for information only.

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* 5. Packaging. Delete paragraph 5 in its entirety and substitute the following:

5. Packaging. The M314A3 Metal Parts shall be packed in accordance with ASTM D3951. It is the responsibility of the contractor to choose packaging that insures the metal parts arrive at their destination site free of physical and mechanical damage and is compatible with the material handling equipment at the destination site.

* 6.2 Delete entire paragraph and substitute the following:

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Packaging requirements (see 5).

c. Requirements for submission of first article sample (see 4.4).

d. Government Furnished Equipment.

e. Requirements for submittal of Acceptance Inspection Equipment.

f. Requirements for ammunition lot numbering and ammo data cards.

* 6.3 Delete the entire paragraph and substitute the following:

6.3 Submission of inspection procedures and equipment designs for approval. Submit equipment designs, as required, to Commander, TACOM-ARDEC, ATTN: AMSTA-AR-QAA-R, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contractor Data Requirements List, DD Form 1423 in the contract.

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* 6.4 Delete the entire paragraph and substitute the following:

6.4 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination the characteristic may be measured or gagged to determine acceptability.

* 6.5 Delete the entire paragraph and substitute the following:

6.5 Visual standards. Visual standards shall be established at contractor plant in compliance with 3.5.4.

* 6.6 Delete the entire paragraph and substitute the following:

6.6 Submission of ballistic test data and ammunition data cards. In addition to the normal distribution of records for items procured by the Department of the Army, two copies of all ballistic test data and ammunition data cards shall be forwarded to Commanding Officer, TACOM-ARDEC, ATTN: AMSTA-AR-QAA-R, Picatinny Arsenal, NJ 07806-5000.

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